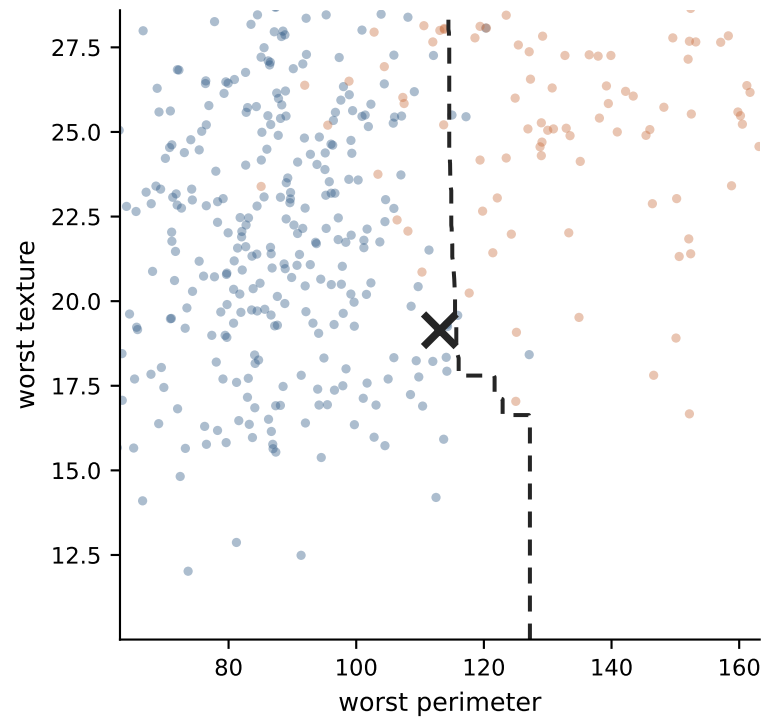


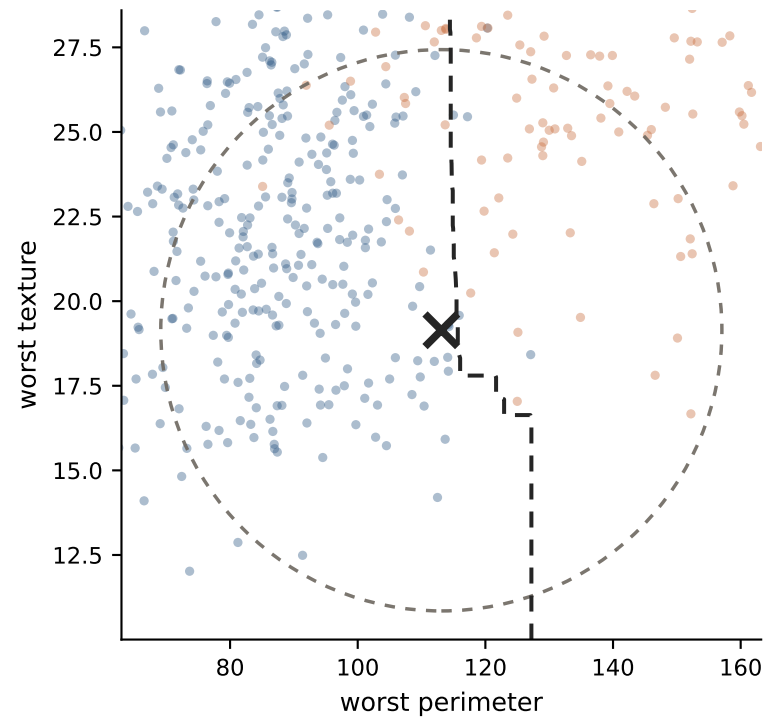
LIME

Local Interpretable Model-agnostic Explanation — patient #67, $P(\text{benign})=0.581$, $R^2=0.36$

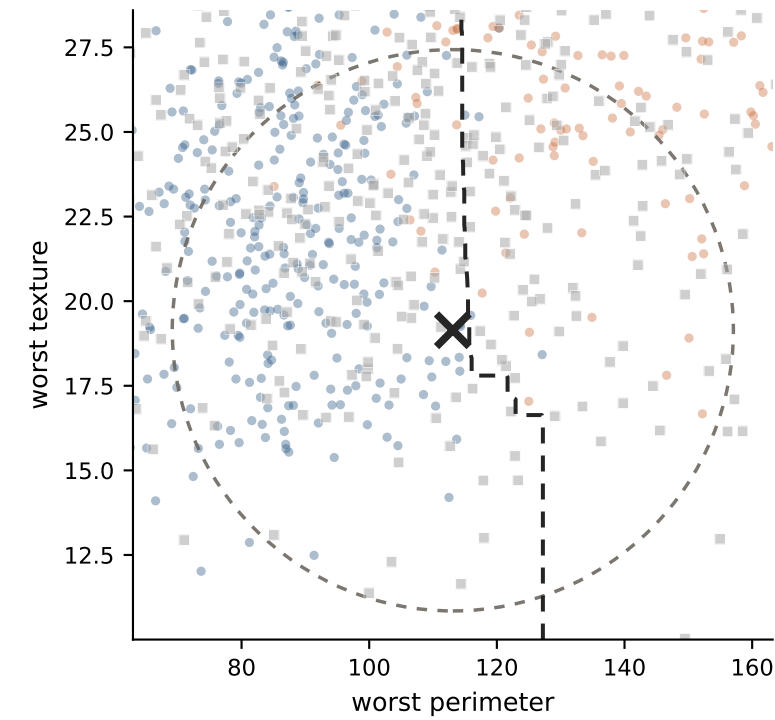
A · the black-box model



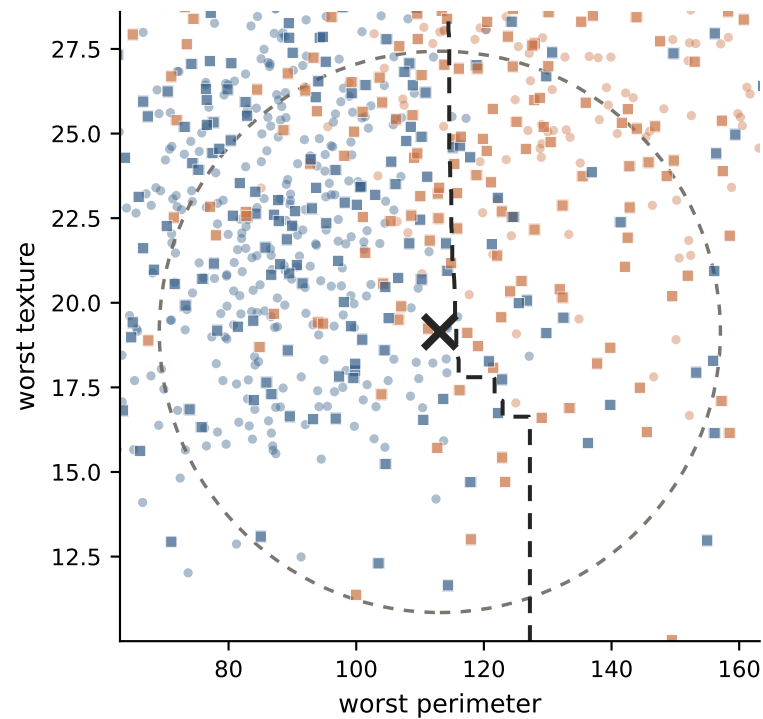
B · the local neighborhood



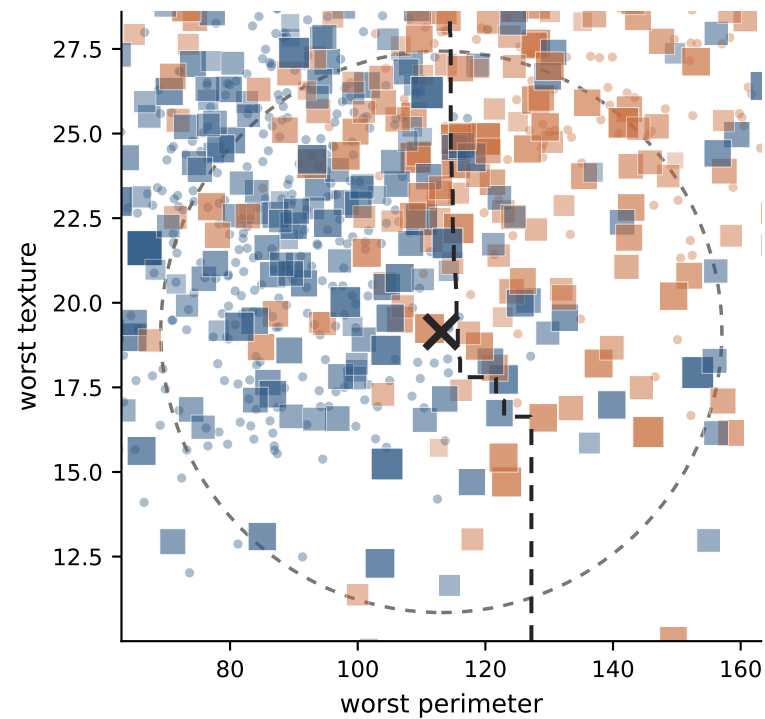
C · perturbation



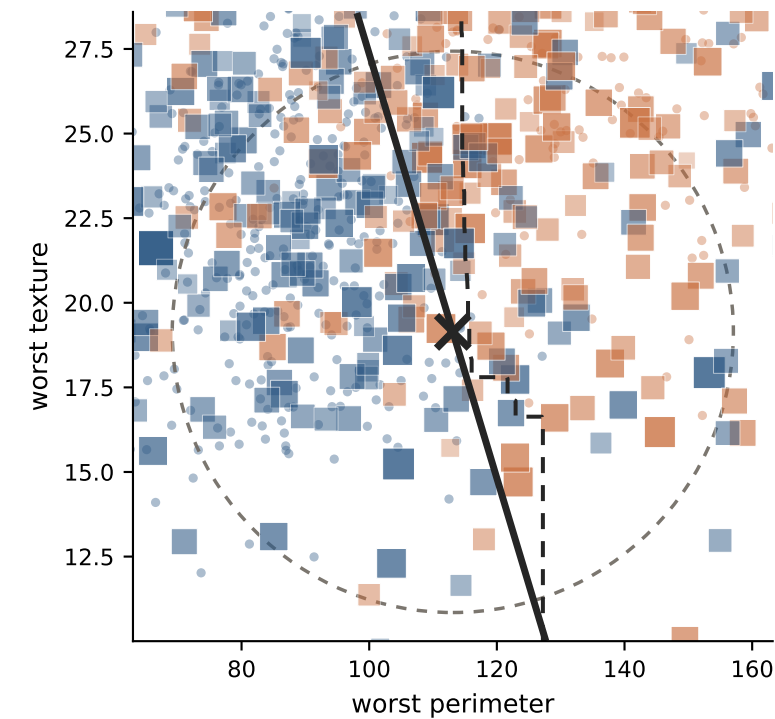
D · predictions on the neighbors



E · proximity weighting



F · the local fit



● real patient · benign ■ neighbor · predicted benign - - - the model's boundary ($P=0.5$) ✕ patient #67
● real patient · malignant ■ neighbor · predicted malignant - - - kernel ring we drew, $w = 0.95$ — local fit (through the patient)